Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C063 **TPK1** 4 genes / 2 species 3 direct pairs	**Y1 TPK1** GenAge 1502 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043 **Y2 TPK2** GenAge 681 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 11000115 **Y3 TPK3** GenAge 1632 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 18340043	No gene in this candidate network	**F1 Pka-C1** GenAge 507 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 17369827	No gene in this candidate network
Direct pairs: F1–Y1 (n:m); F1–Y2 (n:m); F1–Y3 (n:m)				
C108 **unc-51** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 unc-51** GenAge 996 **GenAge annotation** **Unclear** Lifespan: Increase + Decrease Refs: 18219227, 19469880	**F1 Atg1** GenAge 1908 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 25199830	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.